1. Project Overview

The system is an affordable, portable, game-based platform that trains and measures reaction time, peripheral vision, balance and hand-eye coordination for athletes, stroke/injury patients, elderly individuals and children. It combines smart LED targets, IMU and pressure sensors, computer-vision pose tracking and an AI analytics dashboard.

The architecture is delivered in four components:

- **Component 1 – Hardware & Sensor Integration:** ESP32-S3 hub, LED targets, IMU, load cells, FSR, distance sensors.
- **Component 2 – Computer Vision:** OpenCV + MediaPipe for pose, posture and movement detection.
- **Component 3 – Data Processing & Analytics:** filtering, sensor fusion, metrics and ML models.
- **Component 4 – User Interface & Final MVP:** web/mobile dashboard, training games and reports.

2. Hardware Components

Costs are approximate Indian market prices (Robu.in, Robocraze, Amazon India, local electronics markets) and will vary.

2.1 Main Controller

#	Component	Qty	Purpose	Approx. Cost (₹)
1	ESP32-S3 DevKitC-1 (N16R8 – 16 MB flash, 8 MB PSRAM)	1	Main hub: sensor acquisition, LED control, Wi-Fi/BLE, on-board TinyML	650 – 900
2	ESP32-S3 Super Mini / ESP32-C3 Mini (optional)	6 – 8	One per wireless LED target, linked to the hub via ESP-NOW	250 – 350 each
3	XIAO ESP32-S3 Sense (OV2640 camera, mic, SD) or ESP32-S3-EYE (optional)	1	Lightweight on-device vision node; full MediaPipe runs on PC/Pi	1,200 – 1,800

2.2 Sensors

#	Component	Qty	Purpose	Approx. Cost (₹)
4	MPU6050 IMU (upgrade: ICM-20948 / MPU9250)	2 – 4	Wrist / ankle / waist acceleration, angular velocity, orientation	150 – 250 each
5	Load cells – 50 kg half-bridge (set of 4)	1 set	Balance board – weight distribution	400 – 600
6	HX711 24-bit load-cell amplifier	4	One per load cell, so centre of pressure (CoP) can be computed	60 – 100 each
7	FSR402 / FSR406 force-sensitive resistors	4 – 6	Foot pressure zones, hit-force detection on targets	250 – 400 each
8	HC-SR04P (3.3 V) ultrasonic or VL53L0X / VL53L1X ToF	2 – 4	Detects hand/body approach to targets, reach distance	80 – 350 each
9	Arcade push-buttons or capacitive touch pads	6 – 8	Exact hit timestamp for reaction-time measurement	30 – 80 each

#	Component	Qty	Purpose	Approx. Cost (₹)
10	MCP23017 16-bit I²C I/O expander	1	Reads all target buttons with one interrupt pin (frees GPIOs)	120 – 180

2.3 Stimulus & Feedback Outputs

#	Component	Qty	Purpose	Approx. Cost (₹)
11	WS2812B LED rings (12/16 LED) or 1 m strip, 60 LED/m	6 – 8	Smart reaction targets with programmable colour	120 – 200 each
12	74AHCT125 level shifter	1	3.3 V → 5 V for the WS2812B data line	50
13	Active buzzer	2	Audio stimulus / feedback	20 – 30 each
14	Coin vibration motor + 2N2222 / BC547 transistor + 1N4148 diode	2	Haptic feedback (useful for children and patients)	40 – 60 each
15	0.96" OLED (SSD1306, I²C) or 1.3" TFT	1	On-device score / status display	200 – 400

2.4 Computer Vision Hardware (Phase 2)

#	Component	Qty	Purpose	Approx. Cost (₹)
16	USB webcam 1080p, 30 – 60 fps	1	MediaPipe pose and hand tracking	1,200 – 2,500
17	Raspberry Pi 5 (4 GB) + Pi Camera v2, or a laptop	1	Runs the computer-vision pipeline and backend server	6,500 – 8,000 (Pi)

2.5 Power Supply

#	Component	Qty	Purpose	Approx. Cost (₹)
18	5 V 4 A DC adapter (bench setup)	1	Supplies LEDs and electronics	300 – 450
19	18650 Li-ion cells + 2S holder + BMS, or 10,000 mAh power bank	1 set	Portable operation	500 – 1,000
20	LM2596 / MP1584 buck converter (to 5 V)	1 – 2	Regulates battery voltage to 5 V	60 – 100 each
21	1000 µF electrolytic capacitor + 330 Ω resistor	1 each	Protects the WS2812B strip from surges	20
22	TP4056 Li-ion charger module	6 – 8	Charging for wireless target nodes	30 each

2.6 Mechanical & Assembly

#	Component	Qty	Purpose	Approx. Cost (₹)

#	Component	Qty	Purpose	Approx. Cost (₹)
23	Plywood / acrylic plates (≈ 40 × 40 cm, 2 pcs)	1 set	Balance-board top and base	400 – 700
24	3D-printed / acrylic target housings	6 – 8	Enclosures for LED targets	600 – 1,500 total
25	Perfboard / custom PCB, JST connectors, headers, jumper wires	–	Wiring and assembly	400 – 800
26	Velcro straps / wearable bands	2 – 4	Mounting IMUs on wrist and ankle	150
27	MicroSD card module + 16 GB card	1	Offline session logging	80 + 300

2.7 Estimated Budget

Configuration	Includes	Approx. Total (₹)
Core prototype	ESP32-S3 hub, sensors, wired LED targets, power, balance board	6,000 – 9,000
Full MVP	Core + webcam + Raspberry Pi 5 + wireless ESP-NOW target nodes	15,000 – 20,000

3. ESP32-S3 Pin Plan & Design Notes

3.1 Suggested Pin Allocation (DevKitC-1 N16R8)

Function	GPIO	Notes
I²C bus 0 (SDA / SCL)	GPIO 8 / GPIO 9	IMU #1 (0x68), IMU #2 (0x69), OLED (0x3C), MCP23017 (0x20)
I²C bus 1 (SDA / SCL)	GPIO 17 / GPIO 18	IMU #3 (0x68), IMU #4 (0x69)
HX711 DOUT ×4	GPIO 38, 39, 40, 41	Load cells front-left, front-right, back-left, back-right
HX711 SCK (shared)	GPIO 42	Common clock to all four HX711 modules
FSR analog inputs (ADC1)	GPIO 1, 2, 4, 5, 6, 7	Use ADC1 only – ADC2 is unusable while Wi-Fi is on
WS2812B data	GPIO 16 → 74AHCT125 → LEDs	Driven by the RMT peripheral (FastLED / NeoPixelBus)
Ultrasonic #1 TRIG / ECHO	GPIO 11 / GPIO 12	Use HC-SR04P (3.3 V) or a voltage divider on ECHO
Ultrasonic #2 TRIG / ECHO	GPIO 13 / GPIO 14	Same as above
MCP23017 interrupt	GPIO 10	Timestamp button hits inside the ISR
Buzzer / vibration motor	GPIO 21 / GPIO 47	Motor via transistor, never directly

Function	GPIO	Notes
Avoid	GPIO 0, 3, 45, 46 (strapping), 19, 20 (USB), 35 – 37 (PSRAM on N16R8), 43, 44 (UART0), 48 (on-board RGB LED)	

3.2 Design Notes

- **3.3 V logic:** the ESP32-S3 is not 5 V tolerant. Use HC-SR04P or a 1 kΩ/2 kΩ divider on ECHO; drive WS2812B data through the 74AHCT125.
- **Four IMUs without a multiplexer:** the S3 has two I²C controllers and each MPU6050 supports two addresses (AD0 low = 0x68, high = 0x69).
- **Accurate reaction timing:** timestamp stimulus and hit events with esp_timer_get_time() (µs) inside interrupts; update LEDs through RMT so they do not block timing.
- **Analog inputs:** use ADC1 pins only for FSRs, because ADC2 cannot be used while Wi-Fi is active.
- **Power:** WS2812B LEDs can draw up to 60 mA each at full white – size the 5 V supply accordingly and share a common ground with the ESP32-S3.
- **On-device AI:** 8 MB PSRAM allows small TFLite Micro / ESP-DL models (fall detection, balance classification) to run on the board.
- **Native USB:** stream high-rate sensor data straight to the PC for dataset logging.

4. Software Requirements

4.1 Development Environment

Area	Tool / Platform	Notes
Firmware IDE	Arduino IDE 2.x or VS Code + PlatformIO	ESP32 Arduino Core 3.x (or ESP-IDF 5.x)
Vision / AI language	Python 3.10 or 3.11	Versions currently best supported by MediaPipe
Notebook / experiments	Jupyter / Google Colab	Model training and data analysis
Messaging broker	Eclipse Mosquitto (MQTT)	Real-time data link ESP32 ↔ server
Database	SQLite (prototype) → PostgreSQL / Firebase	Users, sessions, trials, raw signals
Frontend	React (web) and/or Flutter (mobile)	Dashboard and training app
Version control	Git + GitHub	Team collaboration
Design / CAD	Fritzing / KiCad, Fusion 360 / Tinkercad	Circuit diagram, PCB, 3D-printed housings

4.2 Firmware (ESP32-S3)

Functional requirements:

- Read IMUs at ≈100 Hz, load cells at 50–80 Hz, FSRs and distance sensors at 50 Hz.

- Generate LED stimuli in random, sequential or programmed patterns with configurable colour, delay and duration.
- Record stimulus, movement-onset and hit timestamps with microsecond resolution; detect false starts.
- Calibrate sensors: load-cell tare and scale, IMU offset zeroing at start-up.
- Transmit data as JSON over MQTT/WebSocket (Wi-Fi) or BLE; buffer to SD card when offline.
- Receive commands from the app: start/stop session, select mode, set difficulty.
- Coordinate wireless targets over ESP-NOW with time synchronisation.

Firmware libraries:

Library	Use
Adafruit_MPU6050 / ElectronicCats MPU6050	IMU readout (DMP optional)
HX711 (bogde) / HX711_ADC	Load-cell readout and tare
FastLED or Adafruit NeoPixel / NeoPixelBus	WS2812B control via RMT
Adafruit_MCP23X17	Button expander with interrupts
NewPing / VL53L0X (Pololu)	Distance sensing
PubSubClient or AsyncMqttClient	MQTT publish/subscribe
ArduinoJson	Packet serialisation
NimBLE-Arduino	Low-power BLE link to mobile app
esp_now (built in)	Wireless target nodes
Adafruit_SSD1306 + GFX	OLED display
TensorFlow Lite Micro / ESP-DL	On-device fall / balance classifier

4.3 Computer Vision Module (Python)

- Capture webcam frames at 30–60 fps with OpenCV.
- Track 33 body landmarks (MediaPipe Pose) and 21 hand landmarks (MediaPipe Hands).
- Compute joint angles, trunk lean, body sway, reach distance and posture symmetry.
- Detect visual movement onset to split reaction time into “react” and “move” phases.
- Detect loss of balance / falls and trigger alerts (safety for elderly and patients).
- Overlay a live skeleton and feedback on the video feed.

4.4 Data Processing & AI Module

- **Filtering:** complementary or Kalman filter for IMU; low-pass / moving average for load cells and FSRs.
- **Sensor fusion:** align IMU, pressure and camera streams by timestamp.
- **Metrics:** reaction time, movement time, accuracy %, misses, false starts, left-vs-right reaction time, CoP sway area and path length, balance stability score, coordination score.
- **ML models:** skill-level classification (Random Forest / XGBoost); fall and instability detection (LSTM / 1D-CNN on IMU windows); progress prediction (regression).
- **Adaptive difficulty:** automatically adjust LED speed, spread and pattern complexity from recent performance.
- **Recommendations:** rule-based + ML suggestions for the next session.

Python libraries:

Library	Use
opencv-python	Camera capture, frame processing, drawing overlays
mediapipe	Pose (33 landmarks) and Hands (21 landmarks) tracking
numpy, pandas, scipy	Signal processing, filtering, feature extraction
filterpy	Kalman filter for IMU fusion
scikit-learn, xgboost	Skill-level classification, regression
tensorflow / keras (export to TFLite)	LSTM / 1D-CNN for fall and instability detection
paho-mqtt	Receive sensor streams from ESP32
fastapi + uvicorn, websockets	REST API and live data to dashboard
sqlalchemy	Database access
matplotlib / plotly	Analysis plots, report charts
reportlab	PDF progress reports for therapists

4.5 Training Modes (Games)

Mode	Description	Target Users
Reaction Drill	Random LED lights up; user hits it as fast as possible	All
Peripheral Vision	User fixes gaze on centre; targets light at the edges	Athletes, patients
Go / No-Go	Hit green targets, ignore red ones	Athletes, children
Static Balance	Hold still on the balance board; sway is measured	Elderly, patients
Weight Shift	Move CoP to on-screen markers	Patients, elderly
Single-Leg Balance	Timed single-leg stance with sway scoring	Athletes
Combined Challenge	Hit targets while balancing on the board	Athletes, advanced users
Colour / Memory Game	Repeat LED sequences with fun visuals and sounds	Children

4.6 Backend, Dashboard & App

- User registration and login with roles: athlete, patient, therapist/coach, parent, admin.
- Live session view showing current stimulus, reaction times, CoP plot and skeleton.
- Progress charts over days/weeks; reaction-time heatmap per target position.
- Comparison with age-group reference values.
- Personalised recommendations and automatic difficulty settings.
- Downloadable PDF report for therapists and coaches; CSV export of raw data.
- REST API (FastAPI) + WebSocket for live data; MQTT bridge to the ESP32-S3.

4.7 Non-Functional Requirements

- **Latency:** stimulus-to-timestamp error below 5 ms; dashboard update below 200 ms.
- **Accuracy:** load-cell error below ±0.5 kg; IMU sampling jitter below 2 ms.
- **Portability:** battery operation of at least 3 hours; setup in under 10 minutes.
- **Safety:** fall alerts, emergency stop, no exposed wiring, rounded target housings.
- **Privacy:** anonymised IDs, consent forms, encrypted storage of health data.
- **Usability:** large, simple UI suitable for children and elderly users.
- **Minimum PC:** Intel i5 / Ryzen 5, 8 GB RAM, USB 3.0, Windows 10/11 or Ubuntu 22.04.

5. Input Datasets

5.1 Public Datasets

These datasets are used to pre-train and validate the pose, fall-detection, activity and balance models before fine-tuning on data collected with the system. Check each dataset's licence and cite it in reports.

Dataset	Source	Data Type	Use in Project
COCO Keypoints	cocodataset.org	Images with 17 body keypoints	Pose-estimation validation, posture models
MPII Human Pose	Max Planck Institute for Informatics	25k images, 16 keypoints, 410 activities	Pose estimation benchmark
UI-PRMD	University of Idaho	Kinect + Vicon skeleton data, 10 rehab exercises, correct vs incorrect	Rehab movement-quality scoring
KIMORE	Università Politecnica delle Marche	RGB-D + skeleton, rehab exercises with clinical scores	Patient exercise assessment
SisFall	University of Antioquia (Sucerquia et al., 2017)	Accelerometer + gyroscope; falls and ADLs incl. elderly subjects	Fall / instability detection from IMU
MobiFall	BMI Lab, TEI Crete	Smartphone IMU data for falls and daily activities	Fall detection
UP-Fall Detection	Universidad Panamericana	Wearable IMU + ambient sensors + cameras	Multimodal (camera + IMU) fall detection
UR Fall Detection	University of Rzeszów	Depth + RGB video and accelerometer	Camera-based fall detection
UCI HAR	UCI ML Repository	Smartphone accelerometer + gyro, 6 activities, 30 subjects	Activity / motion classification
WISDM	UCI ML Repository / Fordham University	Phone and watch IMU data, 18 activities	Activity recognition
Human Balance Evaluation Database	PhysioNet (Santos & Duarte, 2016)	Force-platform CoP data, 163 young & older adults	Balance / sway reference values
NTU RGB+D (60/120)	ROSE Lab, NTU Singapore	3D skeleton + RGB-D, 60–120 actions	Action and movement recognition

Dataset	Source	Data Type	Use in Project
Human3.6M	Ionescu et al.	3.6 M 3D human poses	3D pose / movement analysis

5.2 Custom Dataset (Collected with the System)

No public dataset covers LED-target reaction time combined with balance and pose, so the team must collect its own data using the hardware. Each trial should be logged with the following fields:

Field	Type	Description
user_id	string	Anonymised participant ID
age, gender, group	int / string	Group = athlete / patient / elderly / child
session_id, trial_no	string / int	Session and trial identifiers
mode, difficulty_level	string / int	Training mode and level
target_id, target_angle	int / float	Which LED lit and its position relative to gaze (peripheral analysis)
stimulus_time_us	int	Time the LED turned on (µs)
movement_onset_us	int	First movement detected by IMU / camera
hit_time_us	int	Time the button / FSR registered a hit
reaction_time_ms, movement_time_ms	float	Onset − stimulus, hit − onset
result	enum	hit / miss / false-start / wrong-target
imu_ax … imu_gz (per IMU)	float	6-axis IMU stream @ 100 Hz
lc_fl, lc_fr, lc_bl, lc_br	float	Load cells in kg @ 50–80 Hz
cop_x, cop_y	float	Centre of pressure (computed)
fsr_1 … fsr_6	int	Raw ADC values
distance_cm	float	Ultrasonic / ToF reading
pose_landmarks	array	33 × (x, y, z, visibility) per frame @ 30 fps
label	string	Ground-truth: stable / unstable / fall, skill level

5.3 Data Collection Protocol

- **Participants:** 20–30 volunteers covering athletes, students/children (with parental consent), elderly adults and, if possible, patients via a clinic.
- **Sessions:** 3–5 sessions per person; 30–50 trials per mode per session.
- **Labels:** skill level from baseline tests; stable/unstable/fall from video review; therapist scores where available.
- **Storage:** CSV/Parquet per session plus SQLite/PostgreSQL; video stored only with consent.
- **Split:** 70% training, 15% validation, 15% test, split by participant to avoid data leakage.
- **Ethics:** informed consent, anonymised IDs, institutional ethics approval for patient data.

6. Mapping to Project Phases

Phase	Focus	Hardware	Software	Datasets
1	Hardware & sensors	ESP32-S3, IMU, load cells, FSR, ultrasonic, LEDs	Firmware, MQTT streaming	Raw sensor logs
2	Computer vision	Webcam / Pi camera, Raspberry Pi	OpenCV, MediaPipe	COCO, MPII, UI-PRMD, KIMORE
3	Data processing & AI	—	Filters, fusion, scikit-learn, TensorFlow	SisFall, UP-Fall, UCI HAR, PhysioNet balance, custom
4	UI & final MVP	Enclosures, battery pack	FastAPI, React/Flutter, reports	Custom dataset (validation)